M9.3.b.1 Semantic Web and Ontologies

Tuto/Lab 1: OWL and Reasoning

Part 1: Work environment

1- Download and install the last version of Protégé tool at
https://protege.stanford.edu

Part 2: Importing and inspecting a small ontology

Before starting you may have a look on the following demo
https://youtu.be/4wr25AnDXjs?si=VsMi5rtqFYs9xNZE

1- Download the provided ontology university.ttl available on Celene.
2- Open the file with Protégé.
3- Go through the ontology contents: classes, properties, and individuals. You may also personalize the tool to add for example OntoGraf tab and visualize the graph representation of the ontology.
4- Run a reasoner (Hermit or Pellet) to check ontology consistency and to view new inferred axioms.
5- Correct the inconsistencies and synchronize the reasoner until you get a consistent ontology

Part 3: Ontology enrichement and reasoning

For each of the following steps, add required axioms and then synchronize the reasoner to check the consistency and inspect the newly inferred knowledge and the corresponding explanations.

1- A course should refer to at least one Reading Material that can be Book or MOOC.
Create the class ReadingMaterial and two subclasses Book and MOOC.
Create the property hasReference between Course and ReadingMaterial.
Add new instances of Book, MOOC, and Course and connect them with hasReference consequently.
2- A Book is written by a Person who is an Author whereas a MOOC is created by a ResearchTeam.
Create class Author as subclass of Person and class ResearchTeam.
Create properties writes and creates
Add new instances of classes and connect them with corresponding relations.
3- Now we would like to restrict the previous classes by integrating Topics of courses, books and moocs.
As a first solution, we can model through a property that ranges in Literals coding topic names.
Update the previous classes following this solution.

4- Now, teachers may also be expert for some topics. Create the class ExpertTeacher subsclass of Teacher and having one or more expertise topics.
5- Study the feasibility of defining ExpertTeacher for a specific Course as the author of a book of the same topic as the course.
6- Change the modeling of Topics through the creation of a class Topic and modify the properties consequently.
7- Add a property supervises as inverse property of hasSupervisor.
Study the possibility of inferring hasSupervisor through property composition of properties supervises and worksWith. A Teacher who supervises studentA also supervise studentB whenever studentA works with studentB. You may think about General role inclusion.
8- A teacher may advise a student. Add the property advises correspondingly.
9- Study the possibility of defining a subproperty of advises, helpfullyAdvices corresponding to the case of a teacherA who works with teacherB and advises a student supervised by teacherB.

Part 4: Open reading on used reasoners

1- Read the two scientific papers related to reasoners Pellet and Hermit available on Celene.
    a. Pellet: A practical OWL-DL reasoner
    b. HermiT: An OWL 2 Reasoner